Anomalous Scaling of the Discrete Monitoring Correction for Variance Swaps Under Volterra Stochastic Volatility

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Abstract

We derive the asymptotic behavior of the discrete monitoring correction for variance swaps under a general class of Volterra stochastic volatility models with fractional kernel $K(t) = t^{H-1/2}/\Gamma(H+\frac{1}{2})$, $H \in (0, \frac{1}{2})$. In classical stochastic volatility models, the fair strike of a discretely monitored variance swap converges to its continuous counterpart at rate $O(n^{-1})$, where n is the number of monitoring dates. We show that under Volterra stochastic volatility, the convergence rate deteriorates to $O(n^{-(H+1/2)})$ when the spot–volatility correlation ρ is nonzero. The leading-order coefficient depends on the model only through the averaged quantity $\bar{\varphi} = T^{-1}\int_0^T \mathbb{E}[\sigma_V(V_t)\sqrt{V_t}]\,\mathrm{d}t$, where σ_V is the vol-of-vol function: the anomalous scaling is a *universal* property of the fractional kernel singularity. Rough Heston, rough CIR, and other power-law Volterra models are recovered as special cases.

We prove that the rate $n^{-(H+1/2)}$ is *sharp*: the rescaled gap $n^{H+1/2}\delta(n)$ converges to an explicit nonzero limit, establishing a two-sided bound for all sufficiently large n. A corrected remainder analysis shows that the Malliavin drift correction is $O(n^{-(2H+1)})$—always subdominant to the convexity term $O(n^{-1})$—eliminating a spurious phase transition at $H = \frac{1}{6}$ present in earlier analyses. We characterize the boundary behavior: as $H \to 0^+$, the rate approaches $O(n^{-1/2})$, matching the discrete monitoring bound for jump models; the coefficient is affine in ρ with no blow-up at $\rho = \pm 1$. Numerical experiments with a lifted Markovian approximation confirm the theoretical predictions across a range of Hurst parameters, correlations, and monitoring frequencies.

Keywords: variance swaps, discrete monitoring, rough volatility, Volterra stochastic volatility, Malliavin calculus, leverage effect, fractional kernel, anomalous scaling

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1 Introduction

Variance swaps are among the most liquid volatility derivatives. A variance swap pays the difference between the realized variance of an underlying asset over a fixed period and a

predetermined strike price. In practice, the realized variance is computed from discretely sampled log-returns—typically at daily frequency—while theoretical pricing is based on the continuously monitored quadratic variation.

The gap between the discrete and continuous fair strikes has been studied extensively for classical stochastic volatility models. Broadie and Jain [2008] and Bernard and Cui [2014] derived explicit $O(n^{-1})$ expansions for the Heston and Hull–White models. Bernard et al. [2014] extended the convergence analysis to general time-homogeneous diffusions, establishing that the $O(n^{-1})$ rate is universal among continuous semimartingale volatility models. Jarrow et al. [2013] established an $O(n^{-1/2})$ upper bound for the gap under general semimartingale assumptions—a bound that is tight for jump models but not for continuous ones. A key methodological advance was made by Alòs and Fukasawa [2021], who provided an Edgeworth-type expansion of the discretization error for Itô integrals using Malliavin calculus, with the leading correction expressed through Malliavin derivatives of the integrand. Their framework, however, requires the volatility process to be a semimartingale.

Meanwhile, the rough volatility paradigm [Gatheral et al., 2018, El Euch and Rosenbaum, 2019] has become an established framework for modeling equity volatility. Under rough stochastic volatility, the variance process is driven by a Volterra-type kernel $K(t) = t^{H-1/2}/\Gamma(H+\frac{1}{2})$ with Hurst parameter $H \approx 0.1$, producing path irregularity fundamentally different from classical models. The general theory of Volterra stochastic volatility encompasses rough Heston [El Euch and Rosenbaum, 2019], rough Bergomi [Bayer et al., 2016], and rough CIR-type models [Abi Jaber et al., 2021], all sharing the same fractional kernel singularity. The pricing of discretely monitored variance swaps under rough volatility was addressed numerically by Xi and Wong [2021] using characteristic-function methods, and extended to time-dependent coefficients by Zhuang and Huang [2025]. Both works compute discrete fair strikes without asymptotic analysis of the gap as $n \to \infty$. The simulation of rough Heston itself poses additional challenges addressed by Richard et al. [2023].

Contribution. We make four contributions to the understanding of discrete monitoring corrections under rough volatility.

First, we prove that the discrete monitoring gap under a general class of Volterra stochastic volatility models satisfies

$$K_d(n) - K_c = \frac{C_{\text{lev}}(H)}{n^{H+1/2}} + \frac{C_{\text{eff}}}{n} + o(n^{-1}), \tag{1}$$

where the leverage coefficient depends on the model only through a single averaged quantity:

$$C_{\text{lev}}(H) = \frac{-\rho\,\bar{\varphi}\,T^{H+1/2}}{\Gamma(H+\frac{5}{2})}, \qquad \bar{\varphi} = \frac{1}{T}\int_0^T \mathbb{E}\big[\sigma_V(V_t)\sqrt{V_t}\big]\,\mathrm{d}t, \tag{2}$$

and $C_{\text{eff}} = C_{\text{conv}} + C_{\text{drift}}$ combines the convexity correction with the Malliavin drift correction. For rough Heston with $\sigma_V(v) = \nu\sqrt{v}$, we have $\bar{\varphi} = \nu\bar{V}$ where $\bar{V} = T^{-1}\int_0^T \mathbb{E}[V_t]\,\mathrm{d}t$. The result shows that the anomalous scaling $O(n^{-(H+1/2)})$ is a *universal* property of the fractional kernel singularity and the leverage effect, not a feature specific to the affine structure of rough Heston.

Second, we prove that the rate $n^{-(H+1/2)}$ is *sharp*:

$$\lim_{n\to\infty} n^{H+1/2}\,\delta(n) = C_{\text{lev}}(H) \neq 0, \tag{3}$$

establishing an asymptotic equivalence (not merely an upper bound) and a two-sided inequality for all sufficiently large n.

Third, we correct the remainder analysis. The Malliavin drift correction is $O(n^{-(2H+1)})$, which satisfies $2H + 1 > 1$ for all $H > 0$; hence the drift correction is *always* subdominant to the convexity term $O(n^{-1})$. This eliminates a spurious phase transition at $H = \frac{1}{6}$ that appeared in earlier analyses using a looser bound. The remainder simplifies to $o(n^{-1})$ uniformly in $H \in (0, \frac{1}{2})$.

Fourth, we characterize the boundary behavior of the expansion:

- As $H \to 0^+$: the coefficient remains finite while the exponent approaches $\frac{1}{2}$, matching the $O(n^{-1/2})$ jump-model bound of Jarrow et al. [2013].
- As $\rho \to \pm 1$: the coefficient is affine in ρ with no blow-up; the model reduces to a well-posed one-factor system.
- The sign of C_{lev} determines whether the discrete strike overshoots ($\rho < 0$, equity) or undershoots ($\rho > 0$) the continuous strike.

Our approach extends the Malliavin duality framework of Alòs and Fukasawa [2021] to the Volterra setting, where the variance process is non-semimartingale. The Malliavin derivative $D_u V_s \sim K(s-u)\,\sigma_V(V_u)$ inherits the singularity of the fractional kernel at $s = u$, producing a cross-term $\mathbb{E}[I_i\, M_i^W]$ that decays as $\Delta^{H+3/2}$ per monitoring interval rather than the Δ^2 of smooth models.

Our result fills a gap between the numerical studies of Xi and Wong [2021] and Zhuang and Huang [2025] (who computed discrete strikes without asymptotic analysis) and the general convergence framework of Jarrow et al. [2013]. We emphasize that these address different questions: Jarrow et al. establish an $O(n^{-1/2})$ *upper bound* under general semimartingale assumptions (tight for jump models), while we derive the *exact rate* in the Volterra SV class. The log-price $X = \ln S$ in Volterra SV models is itself a semimartingale, so their bound applies; our Malliavin-based computation sharpens it from $O(n^{-1/2})$ to the exact $O(n^{-(H+1/2)})$.

Related literature. The theory of Volterra stochastic volatility models has developed rapidly. Abi Jaber et al. [2021] established existence, uniqueness, and moment bounds for affine Volterra processes; see also Abi Jaber and El Euch [2019] for lifted Markovian approximations. Malliavin calculus for Volterra processes was developed by Alòs et al. [2007] and Nualart [2006], with recent extensions by Viens and Zhang [2019]. The short-time asymptotic behavior of implied volatility under rough models was studied by Forde et al. [2022] and Friz et al. [2022]. Fukasawa [2021] derived the asymptotic behavior of realized variance under microstructure noise with rough volatility. Our work complements these by characterizing the *monitoring frequency* effect, which is distinct from the microstructure effect.

Outline. Section 2 introduces the Volterra SV model class and defines the discrete monitoring gap. Section 3 states and proves the main result for rough Heston as a motivating special case. Section 4 extends the result to general Volterra SV. Section 5 establishes sharpness of the exponent. Section 6 corrects the remainder analysis. Section 7 characterizes boundary behavior. Section 8 presents Monte Carlo validation. Section 9 discusses practical implications. Section 11 concludes. Appendices provide complete proofs of the error bounds and Malliavin differentiability.

Notation. Throughout, $K(t) = t^{H-1/2}/\Gamma(H+\frac{1}{2})$ denotes the Riemann–Liouville fractional kernel, $\mathbb{D}^{1,2}$ the Malliavin Sobolev space, D_u the Malliavin derivative with respect to W_u, and $\Delta = T/n$ the monitoring interval. Constants C, c, C' are generic and may change from line to line; named constants ($C_{\mathrm{lev}}, C_{\mathrm{conv}}, C_R$, etc.) are fixed. We write $f(n) = O(g(n))$ if $|f(n)| \le C\, g(n)$ for all $n \ge n_0$, and $f(n) = o(g(n))$ if $f(n)/g(n) \to 0$.

2 Setup

2.1 Volterra stochastic volatility models

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\ge 0}, \mathbb{Q})$ carrying two independent standard Brownian motions W and B.

Definition 1 (Volterra SV model)**.** A *Volterra stochastic volatility model* with Hurst parameter $H \in (0, \frac{1}{2})$ is a pair (S, V) satisfying

$$\mathrm{d}S_t = \sqrt{V_t}\, S_t\big(\rho\, \mathrm{d}W_t + \bar{\rho}\, \mathrm{d}B_t\big), \quad S_0 > 0, \tag{4}$$

$$V_t = V_0 + \int_0^t K(t-s)\, b(V_s)\, \mathrm{d}s + \int_0^t K(t-s)\, \sigma_V(V_s)\, \mathrm{d}W_s, \tag{5}$$

where $\bar{\rho} = \sqrt{1-\rho^2}$, $K(t) = t^{H-1/2}/\Gamma(H+\frac{1}{2})$ is the fractional kernel, $\rho \in [-1, 1]$ is the leverage parameter, $b\colon \mathbb{R}_+ \to \mathbb{R}$ is the drift function, $\sigma_V\colon \mathbb{R}_+ \to \mathbb{R}_+$ is the vol-of-vol function, and $V_0 > 0$ is the initial variance.

The fractional kernel satisfies the fundamental convolution identity

$$\int_0^t K(t-s)\, K(s)\, \mathrm{d}s = \frac{t^{2H}}{\Gamma(2H+1)}, \tag{6}$$

which follows from the Beta function integral $B(H+\frac{1}{2}, H+\frac{1}{2}) = \Gamma(H+\frac{1}{2})^2/\Gamma(2H+1)$. This identity plays a central role in bounding the Malliavin derivative remainder (Section 6).

Assumption 1 (Standing assumptions)**.** *Throughout the paper, we assume:*

(A1) ***Integrability:*** $\mathbb{E}\big[\int_0^T V_t^2\, \mathrm{d}t\big] < \infty$.

(A2) ***Malliavin differentiability:*** $I_i = \int_{t_i}^{t_i+\Delta} V_s\, \mathrm{d}s \in \mathbb{D}^{1,2}$ *for each* i.

(A3) ***Lipschitz drift:*** b *is Lipschitz continuous with constant* L_b.

(A4) ***Vol-of-vol regularity:*** σ_V *is locally Lipschitz on* $(0,\infty)$ *with* $\sigma_V(v) > 0$ *for* $v > 0$, *and* $\bar{\varphi}_{\max} := \sup_{t\in[0,T]} \mathbb{E}[\sigma_V(V_t)\sqrt{V_t}] < \infty$.

(A5) ***Short-range covariance:*** $\mathrm{Cov}(V_s, V_u) = O(|s-u|^{2H})$ *as* $|s-u| \to 0$, *uniformly in* $s, u \in [0, T]$.

Remark 1 (Verification of the standing assumptions)**.** *For* rough Heston *($b(v) = \theta - \kappa v$, $\sigma_V(v) = \nu\sqrt{v}$), assumptions* (A1), (A3)–(A5) *are established in El Euch and Rosenbaum [2019] under standard parameter conditions ($\kappa > 0$, $\theta > 0$, $\nu > 0$, $V_0 > 0$). Assumption* (A2) *is verified in Appendix E via a localization argument following Alòs et al. [2007]. For* affine Volterra processes *more generally, Abi Jaber et al. [2021] provide the existence, uniqueness, and moment theory. For non-affine Volterra SDEs with Lipschitz coefficients,* (A1) *and* (A5) *follow from standard Volterra resolvent estimates; see, e.g., Gripenberg et al. [1990].*

2.2 Special cases

The Volterra SV framework encompasses several important models.

Example 1 (Rough Heston)**.** *$b(v) = \theta - \kappa v$, $\sigma_V(v) = \nu\sqrt{v}$ with $\kappa > 0$, $\theta > 0$, $\nu > 0$. The drift is Lipschitz with $L_b = \kappa$, and $\sigma_V(v)\sqrt{v} = \nu v$, so $\bar{\varphi} = \nu\bar{V}$ where $\bar{V} = T^{-1}\int_0^T \mathbb{E}[V_t]\,\mathrm{d}t$.*

Example 2 (Rough CIR with general power)**.** *$b(v) = \theta - \kappa v$, $\sigma_V(v) = \nu\, v^\gamma$ with $\gamma \in [\frac{1}{2}, 1]$. Then $\sigma_V(v)\sqrt{v} = \nu\, v^{\gamma+1/2}$, so $\bar{\varphi} = \nu \cdot T^{-1}\int_0^T \mathbb{E}[V_t^{\gamma+1/2}]\,\mathrm{d}t$. The case $\gamma = \frac{1}{2}$ recovers rough Heston.*

Example 3 (Mean-reverting SABR-type)**.** *$b(v) = \theta - \kappa v$, $\sigma_V(v) = \nu\, v$ (log-normal vol-of-vol). Then $\sigma_V(v)\sqrt{v} = \nu\, v^{3/2}$, so $\bar{\varphi} = \nu \cdot T^{-1}\int_0^T \mathbb{E}[V_t^{3/2}]\,\mathrm{d}t$.*

Remark 2 (Rough Bergomi)**.** *The rough Bergomi model [Bayer et al., 2016] has $V_t = \xi(t)\exp(\eta W_t^H - \frac{1}{2}\eta^2 t^{2H})$ where W^H is a fractional Brownian motion with Hurst parameter H. This is a Gaussian Volterra process,* not *a Volterra SDE of the form* (5)*. Nevertheless, its Malliavin derivative satisfies $D_u V_t = \eta\, K(t-u)\, V_t$, sharing the same kernel singularity. The leading-order analysis therefore carries over with $\bar{\varphi} = \eta \cdot T^{-1}\int_0^T \mathbb{E}[V_t^{3/2}]\,\mathrm{d}t$. A complete treatment requires extending the Clark–Ocone argument to the Gaussian Volterra setting, which we leave to future work.*

2.3 Variance swap fair strikes

The *continuous* variance swap fair strike is

$$K_c = \frac{1}{T}\mathbb{E}\Big[\int_0^T V_t\,\mathrm{d}t\Big] = \bar{V}, \tag{7}$$

where $\bar{V} = T^{-1}\int_0^T \mathbb{E}[V_t]\,\mathrm{d}t$.

For n equally spaced monitoring dates $t_i = iT/n$, $i = 0, \ldots, n$, the *discrete* variance swap fair strike is

$$K_d(n) = \frac{1}{T}\sum_{i=0}^{n-1}\mathbb{E}\Big[\Big(\ln\frac{S_{t_{i+1}}}{S_{t_i}}\Big)^2\Big]. \tag{8}$$

Definition 2 (Discrete monitoring gap)**.** The *discrete monitoring gap* is $\delta(n) := K_d(n) - K_c$.

We write $\Delta = T/n$ for the monitoring interval length.

3 Main result: the rough Heston case

We first state and prove the result for rough Heston (Example 1), then generalize in Section 4.

3.1 Decomposition of the gap

On the i-th interval $[t_i, t_i+\Delta]$, define

$$I_i = \int_{t_i}^{t_i+\Delta} V_s \,\mathrm{d}s, \qquad M_i^W = \int_{t_i}^{t_i+\Delta} \sqrt{V_s}\,\mathrm{d}W_s, \qquad M_i^B = \int_{t_i}^{t_i+\Delta} \sqrt{V_s}\,\mathrm{d}B_s. \tag{9}$$

From (4) the log-return is $\Delta X_i := \ln(S_{t_{i+1}}/S_{t_i}) = -\frac{1}{2} I_i + \rho M_i^W + \bar{\rho}\, M_i^B$.

Lemma 3 (Gap decomposition)**.** *Under Assumption* (A1)*,*

$$\delta(n) = \frac{1}{T}\sum_{i=0}^{n-1}\Big[\frac{1}{4}\,\mathbb{E}\big[I_i^2\big] - \rho\,\mathbb{E}\big[I_i\, M_i^W\big]\Big]. \tag{10}$$

Proof. Expanding $(\Delta X_i)^2$:

$$\begin{aligned}\mathbb{E}[(\Delta X_i)^2] = {}& \tfrac{1}{4}\,\mathbb{E}[I_i^2] + \rho^2\,\mathbb{E}[(M_i^W)^2] + \bar{\rho}^2\,\mathbb{E}[(M_i^B)^2] \\ & - \rho\,\mathbb{E}[I_i\, M_i^W] - \bar{\rho}\,\mathbb{E}[I_i\, M_i^B] + 2\rho\bar{\rho}\,\mathbb{E}[M_i^W M_i^B].\end{aligned}$$

By the Itô isometry, $\mathbb{E}[(M_i^W)^2] = \mathbb{E}[(M_i^B)^2] = \mathbb{E}[I_i]$. Since V is adapted to $\sigma(W)$ and B is independent of W, we have $\mathbb{E}[I_i\, M_i^B] = 0$ and $\mathbb{E}[M_i^W M_i^B] = 0$. Therefore

$$\mathbb{E}[(\Delta X_i)^2] = \mathbb{E}[I_i] + \tfrac{1}{4}\mathbb{E}[I_i^2] - \rho\,\mathbb{E}[I_i\, M_i^W].$$

Summing over i and subtracting $K_c = T^{-1}\sum \mathbb{E}[I_i]$ gives (10). □

3.2 The convexity term

Lemma 4 (Convexity contribution)**.** *Under Assumptions* (A1) *and* (A5)*,*

$$\frac{1}{4T}\sum_{i=0}^{n-1}\mathbb{E}[I_i^2] = \frac{C_{\mathrm{conv}}}{n} + O(n^{-(1+2H)}), \tag{11}$$

where $C_{\mathrm{conv}} = \frac{1}{4T}\int_0^T \mathbb{E}[V_t]^2\,\mathrm{d}t$.

Proof. Write $\mathbb{E}[I_i^2] = (\mathbb{E}[I_i])^2 + \mathrm{Var}(I_i)$.

First moment squared. $(\mathbb{E}[I_i])^2 = \big(\int_{t_i}^{t_i+\Delta}\mathbb{E}[V_s]\,\mathrm{d}s\big)^2$. Since $t \mapsto \mathbb{E}[V_t]$ is continuous on $[0,T]$, Taylor expansion gives $\int_{t_i}^{t_i+\Delta}\mathbb{E}[V_s]\,\mathrm{d}s = \mathbb{E}[V_{t_i}]\,\Delta + O(\Delta^2)$, so $(\mathbb{E}[I_i])^2 = \mathbb{E}[V_{t_i}]^2\,\Delta^2 + O(\Delta^3)$. Summing:

$$\sum_{i=0}^{n-1}(\mathbb{E}[I_i])^2 = \Delta^2\sum_{i=0}^{n-1}\mathbb{E}[V_{t_i}]^2 + O(n\Delta^3) = \Delta\Big(\int_0^T \mathbb{E}[V_t]^2\,\mathrm{d}t + O(\Delta)\Big).$$

Hence $(4T)^{-1}\sum(\mathbb{E}[I_i])^2 = C_{\text{conv}}/n + O(n^{-2})$.

Variance term. $\mathrm{Var}(I_i) = \int_{t_i}^{t_i+\Delta}\int_{t_i}^{t_i+\Delta} \mathrm{Cov}(V_s, V_u)\,\mathrm{d}s\,\mathrm{d}u$. Under Assumption (A5), $\mathrm{Cov}(V_s, V_u) = O(|s-u|^{2H})$. Therefore

$$\mathrm{Var}(I_i) \le C\int_0^\Delta\int_0^\Delta |s-u|^{2H}\,\mathrm{d}s\,\mathrm{d}u = C'\Delta^{2+2H},$$

and $\sum_i \mathrm{Var}(I_i) = O(n\,\Delta^{2+2H}) = O(\Delta^{1+2H}) = O(n^{-(1+2H)})$. □

3.3 The leverage term

This is the key contribution. We use the Clark–Ocone formula from Malliavin calculus.

Proposition 5 (Leverage contribution—rough Heston)**.** *Under Assumptions* (A1)–(A4) *with* $b(v) = \theta - \kappa v$ *and* $\sigma_V(v) = \nu\sqrt{v}$*,*

$$-\frac{\rho}{T}\sum_{i=0}^{n-1}\mathbb{E}[I_i\,M_i^W] = \frac{C_{\text{lev}}^{\text{rH}}(H)}{n^{H+1/2}} + O\big(n^{-(2H+1)}\big), \tag{12}$$

where

$$C_{\text{lev}}^{\text{rH}}(H) = \frac{-\rho\,\nu\,\bar{V}\,T^{H+1/2}}{\Gamma\big(H+\frac{5}{2}\big)}. \tag{13}$$

Proof. **Step 1 (Clark–Ocone duality).** By the Clark–Ocone formula [Nualart, 2006],

$$\mathbb{E}[I_i\,M_i^W] = \mathbb{E}\int_{t_i}^{t_i+\Delta} D_u I_i\,\sqrt{V_u}\,\mathrm{d}u, \tag{14}$$

where $D_u I_i = \int_u^{t_i+\Delta} D_u V_s\,\mathrm{d}s$ for $u \in [t_i, t_i+\Delta]$.

Step 2 (Leading-order Malliavin derivative). The Volterra SDE (5) with $\sigma_V(v) = \nu\sqrt{v}$ yields (see Appendix A)

$$D_u V_s = K(s-u)\,\nu\sqrt{V_u} + R(u,s), \qquad s > u, \tag{15}$$

where the remainder R satisfies

$$|R(u,s)| \le \frac{\kappa\,\nu\sqrt{V_u}}{\Gamma(2H+1)}\,(s-u)^{2H}. \tag{16}$$

The bound (16) is established in Appendix B.

Step 3 (Inner integral).

$$\int_u^{t_i+\Delta} K(s-u)\,\mathrm{d}s = \frac{1}{\Gamma(H+\frac{1}{2})}\int_0^{t_i+\Delta-u}\tau^{H-1/2}\,\mathrm{d}\tau = \frac{(t_i+\Delta-u)^{H+1/2}}{\Gamma(H+\frac{3}{2})}. \tag{17}$$

Step 4 (Substitution). At leading order,

$$\mathbb{E}[I_i\,M_i^W] = \nu\,\mathbb{E}\int_{t_i}^{t_i+\Delta} V_u\cdot\frac{(t_i+\Delta-u)^{H+1/2}}{\Gamma(H+\frac{3}{2})}\,\mathrm{d}u + E_i,$$

where E_i is the contribution from the remainder R. Since $\mathbb{E}[V_u]$ is continuous, replacing $\mathbb{E}[V_u] \approx \mathbb{E}[V_{t_i}]$ and substituting $w = t_i + \Delta - u$:

$$\mathbb{E}[I_i\, M_i^W] = \frac{\nu\, \mathbb{E}[V_{t_i}]\, \Delta^{H+3/2}}{\Gamma(H+\frac{5}{2})} + E_i + O(\Delta^{H+5/2}), \tag{18}$$

where the $O(\Delta^{H+5/2})$ term accounts for the approximation $\mathbb{E}[V_u] \approx \mathbb{E}[V_{t_i}]$.

Step 5 (Summation). Summing over i:

$$-\frac{\rho}{T}\sum_{i=0}^{n-1}\mathbb{E}[I_i\, M_i^W] = \frac{-\rho\,\nu\,\Delta^{H+3/2}}{T\,\Gamma(H+\frac{5}{2})}\sum_{i=0}^{n-1}\mathbb{E}[V_{t_i}] - \frac{\rho}{T}\sum_{i=0}^{n-1}E_i + O(n^{-(H+3/2)}). \tag{19}$$

Since $\sum_i \mathbb{E}[V_{t_i}] = n\bar{V} + O(1)$ (Riemann sum approximation) and $\Delta^{H+3/2}\cdot n = T^{H+3/2}/n^{H+1/2}$, the leading term gives $C_{\text{lev}}^{\text{rH}}(H)/n^{H+1/2}$.

For the remainder sum, we show in Appendix D that $\sum_i |E_i| = O(n^{-(2H+1)} \cdot T)$, establishing (12). □

Combining Lemma 3, Lemma 4, and Proposition 5:

Theorem 6 (Rough Heston discrete monitoring gap)**.** *Under Assumptions* (A1)–(A4) *with rough Heston dynamics, the discrete monitoring gap satisfies*

$$\delta(n) = \frac{C_{\text{lev}}^{\text{rH}}(H)}{n^{H+1/2}} + \frac{C_{\text{conv}}}{n} + o(n^{-1}), \tag{20}$$

with $C_{\text{lev}}^{\text{rH}}$ as in (13) *and C_{conv} as in Lemma 4. In particular:*

(i) For $H < \frac{1}{2}$ and $\rho \neq 0$, the leading term is $O(n^{-(H+1/2)})$ with $H + \frac{1}{2} < 1$.

(ii) For $\rho = 0$, the leverage term vanishes and $\delta(n) = C_{\text{conv}}/n + o(n^{-1})$.

(iii) For $H = \frac{1}{2}$ (classical Heston), both terms are $O(n^{-1})$ and the result is consistent with Bernard and Cui [2014].

Remark 3 (Comparison with Alòs and Fukasawa [2021])**.** *The Edgeworth expansion of Alòs and Fukasawa [2021] applies to discretization errors of Itô integrals $\sum_i f(X_{t_i})(X_{t_{i+1}} - X_{t_i}) - \int_0^T f(X_t)\,\mathrm{d}X_t$ when the integrand is a semimartingale. Their leading correction involves $D_u f(X_t)$, which for smooth (semimartingale) volatility is $O(1)$, giving an $O(n^{-1})$ correction. Our setting differs in two ways: (i) the variance process V in the Volterra model is* not *a semimartingale (its paths have Hölder regularity $H < \frac{1}{2}$), so their framework does not directly apply; and (ii) the Malliavin derivative $D_u V_s$ is singular at $s = u$ (blowing up as $(s-u)^{H-1/2}$), which produces the anomalous exponent. Our approach can be viewed as an extension of their duality technique to the singular-kernel setting, with the key new ingredient being the Beta convolution identity* (6) *that controls the Picard iteration for the Malliavin derivative.*

Remark 4 (Finite-n effective exponent)**.** *Over a finite range $n \in [n_{\min}, n_{\max}]$, log-log regression of $\delta(n)$ yields an effective exponent $\hat{\alpha}$ strictly between $H + \frac{1}{2}$ and 1. This is because the sub-leading $O(n^{-1})$ convexity term contributes relatively more at small n, steepening the log-log slope toward 1. Concretely, if $\delta(n) \approx An^{-(H+1/2)} + Bn^{-1}$ with $A, B > 0$, the local exponent $\alpha_{\text{loc}}(n) = -\partial \log \delta / \partial \log n$ satisfies $H + \frac{1}{2} < \alpha_{\text{loc}}(n) < 1$ for all finite n, and $\alpha_{\text{loc}}(n) \to H + \frac{1}{2}$ as $n \to \infty$.*

4 Extension to general Volterra stochastic volatility

4.1 The model-dependent quantity $\bar{\varphi}$

The key observation is that the proof of Proposition 5 uses the rough Heston structure *only* in one place: the specific form of the Malliavin derivative leading term $D_u V_s \approx K(s-u)\,\nu\sqrt{V_u}$. For general Volterra SV (5), the leading-order Malliavin derivative is

$$D_u V_s = K(s-u)\,\sigma_V(V_u) + R(u,s), \tag{21}$$

where σ_V replaces $\nu\sqrt{\cdot}$. The cross-term that drives the leverage contribution becomes

$$\mathbb{E}\int_{t_i}^{t_i+\Delta} \big[K(s-u)\,\sigma_V(V_u)\big]\,\sqrt{V_u}\,\mathrm{d}u \;\rightsquigarrow\; \mathbb{E}\big[\sigma_V(V_u)\sqrt{V_u}\big], \tag{22}$$

which motivates the definition:

Definition 7 (Leverage sensitivity function)**.** The *leverage sensitivity function* is $\varphi(t) := \mathbb{E}[\sigma_V(V_t)\sqrt{V_t}]$, and its time average is $\bar{\varphi} := T^{-1}\int_0^T \varphi(t)\,\mathrm{d}t$.

Remark 5 (Interpretation of $\bar{\varphi}$)**.** *The quantity $\varphi(t)$ measures the expected instantaneous contribution of the vol-of-vol to the leverage cross-term at time t. For rough Heston, $\varphi(t) = \nu\,\mathbb{E}[V_t]$, and $\bar{\varphi} = \nu\bar{V}$ is proportional to the average variance. For rough CIR with $\sigma_V(v) = \nu v^\gamma$, $\varphi(t) = \nu\,\mathbb{E}[V_t^{\gamma+1/2}]$, and $\bar{\varphi}$ involves the $(\gamma+\frac{1}{2})$-th moment of V. The general formulation shows that what matters for discrete monitoring is not the vol-of-vol function σ_V per se, but its interaction with the spot–volatility covariance through the square-root weighting $\sqrt{V_u}$ inherited from the Clark–Ocone formula.*

4.2 Step-by-step proof audit

We audit each step of the proof to identify what uses model-specific structure.

Step 1: Gap decomposition (Lemma 3). Uses only: the log-return structure (4), Itô isometry, and independence of W and B. *No model specificity.*

Step 2: Convexity term (Lemma 4). Uses the short-range covariance $\mathrm{Cov}(V_s, V_u) = O(|s-u|^{2H})$. This holds for any Volterra SV model with kernel K under Assumption (A5), which follows from Lipschitz conditions on b and σ_V combined with the Volterra resolvent theory [cf. Gripenberg et al., 1990, Abi Jaber et al., 2021]. *General: no affine structure needed.*

Step 3: Clark–Ocone duality (Step 1). Uses $I_i \in \mathbb{D}^{1,2}$ and the Brownian representation. *General under Assumption* (A2).

Step 4: Leading Malliavin derivative (Step 2). For general Volterra SDE (5), the Malliavin derivative satisfies the linear Volterra equation

$$D_u V_t = K(t-u)\,\sigma_V(V_u) + \int_u^t K(t-s)\,b'(V_s)\,D_u V_s\,\mathrm{d}s + \int_u^t K(t-s)\,\sigma_V'(V_s)\,D_u V_s\,\mathrm{d}W_s. \tag{23}$$

The leading order is $K(t-u)\sigma_V(V_u)$; the drift correction involves b' (bounded under Lipschitz assumption on b); the stochastic correction vanishes in expectation (martingale property, see Appendix C). *General under Assumptions* (A3)–(A4).

Step 5: Inner integral and summation (Steps 3–5). Uses only the kernel integration formula $\int_0^\Delta K(\tau)\,\mathrm{d}\tau = \Delta^{H+1/2}/\Gamma(H+\frac{3}{2})$ and the Riemann sum approximation $\sum \mathbb{E}[f(V_{t_i})] \approx n \cdot T^{-1}\int \mathbb{E}[f(V_t)]\,\mathrm{d}t$. *General.*

Remark 6 (Which assumptions can be relaxed)**.** *The standing assumptions can be partially relaxed:*

(i) (A3) *(Lipschitz b) is used only for the drift correction bound* (40)*. If b is merely locally Lipschitz on $(0,\infty)$ (e.g., $b(v) = \theta v^{-\alpha} - \kappa v$ for some $\alpha > 0$), the bound still holds on the event $\{V_t \geq \epsilon\}$ with localization.*

(ii) (A4) *requires $\bar{\varphi}_{\max} < \infty$. For $\sigma_V(v) = \nu v^\gamma$, this needs $\sup_t \mathbb{E}[V_t^{\gamma+1/2}] < \infty$, which holds when the moment of order $\gamma + \frac{1}{2}$ exists. For rough Heston ($\gamma = \frac{1}{2}$), this is simply $\sup_t \mathbb{E}[V_t] < \infty$, guaranteed by mean-reversion. For the log-normal case ($\gamma = 1$), one needs $\sup_t \mathbb{E}[V_t^{3/2}] < \infty$, which requires stronger moment conditions.*

(iii) (A5) *(short-range covariance) follows automatically from the Volterra structure under* (A3)–(A4) *for any $H \in (0, \frac{1}{2})$. The Hölder exponent of $t \mapsto \mathbb{E}[V_t^2]$ is at least $2H$, which gives the desired covariance decay. For affine Volterra processes, the exact covariance $\mathrm{Cov}(V_s, V_u) = R(s,u)$ satisfies a Volterra integral equation of the second kind with kernel K, and the solution inherits the singularity exponent of K [El Euch and Rosenbaum, 2019, Proposition 3.1].*

4.3 The general theorem

Theorem 8 (General Volterra SV discrete monitoring gap)**.** *Under Assumptions* (A1)–(A5)*, the discrete monitoring gap satisfies*

$$\delta(n) = \frac{C_{\mathrm{lev}}(H)}{n^{H+1/2}} + \frac{C_{\mathrm{eff}}}{n} + o(n^{-1}), \tag{24}$$

where

$$C_{\mathrm{lev}}(H) = \frac{-\rho\,\bar{\varphi}\,T^{H+1/2}}{\Gamma(H+\frac{5}{2})}, \qquad \bar{\varphi} = \frac{1}{T}\int_0^T \mathbb{E}\big[\sigma_V(V_t)\sqrt{V_t}\big]\,\mathrm{d}t, \tag{25}$$

and

$$C_{\mathrm{eff}} = C_{\mathrm{conv}} + C_{\mathrm{drift}}, \tag{26}$$

with $C_{\mathrm{conv}} = (4T)^{-1}\int_0^T \mathbb{E}[V_t]^2\,\mathrm{d}t$ and C_{drift} given explicitly in (33)*. In particular:*

(i) For $H < \frac{1}{2}$ and $\rho \neq 0$ and $\bar{\varphi} \neq 0$, the leading term is $O(n^{-(H+1/2)})$.

(ii) For $\rho = 0$ or $\bar{\varphi} = 0$, $\delta(n) = C_{\mathrm{eff}}/n + o(n^{-1})$.

Proof. The gap decomposition (Lemma 3) and convexity estimate (Lemma 4) carry over unchanged. For the leverage term, the leading-order Malliavin derivative of the general Volterra SDE is $D_u V_s = K(s-u)\,\sigma_V(V_u) + R(u,s)$, where the remainder satisfies $|R(u,s)| \le C\,\sqrt{V_u}\,(s-u)^{2H}$ under Assumptions (A3)–(A4) (Appendix B, with κ replaced by L_b).

The stochastic correction in (23) vanishes in expectation when computing $\mathbb{E}[I_i M_i^W]$: the inner stochastic integral from u to s is a martingale, and the tower property $\mathbb{E}[V_u \cdot \mathbb{E}[\cdot \mid \mathcal{F}_u]] = 0$ eliminates it (Appendix C).

Substituting into the Clark–Ocone formula and repeating the integration of Proposition 5 with $\sigma_V(V_u)\sqrt{V_u}$ in place of νV_u:

$$\begin{aligned}\mathbb{E}[I_i\, M_i^W] &= \frac{1}{\Gamma(H+\frac{3}{2})}\int_{t_i}^{t_i+\Delta} (t_i+\Delta-u)^{H+1/2}\,\mathbb{E}\big[\sigma_V(V_u)\,V_u\big]\,\mathrm{d}u + E_i \\ &= \frac{\varphi(t_i)\,\Delta^{H+3/2}}{\Gamma(H+\frac{5}{2})} + E_i + O(\Delta^{H+5/2}),\end{aligned}$$

where $\varphi(t_i) = \mathbb{E}[\sigma_V(V_{t_i})\sqrt{V_{t_i}}]$. Summing:

$$-\frac{\rho}{T}\sum_{i=0}^{n-1}\mathbb{E}[I_i M_i^W] = \frac{-\rho\,\bar\varphi\,T^{H+1/2}}{\Gamma(H+\frac{5}{2})\,n^{H+1/2}} + O(n^{-(2H+1)}),$$

using $\sum_i \varphi(t_i) = n\bar\varphi + O(1)$ and the remainder bound from Appendix D. Combining with the convexity term gives (24). □

Corollary 9 (Recovery of special cases)**.**

(a) ***Rough Heston:*** *$\sigma_V(v) = \nu\sqrt{v}$, so $\bar\varphi = \nu\bar V$ and $C_{\mathrm{lev}} = -\rho\nu\bar V\,T^{H+1/2}/\Gamma(H+\frac{5}{2})$, recovering Theorem 6.*

(b) ***Rough CIR (γ-power):*** *$\sigma_V(v) = \nu v^\gamma$, $\gamma \in [\frac{1}{2}, 1]$, so $\bar\varphi = \nu \cdot T^{-1}\int_0^T \mathbb{E}[V_t^{\gamma+1/2}]\,\mathrm{d}t$. The anomalous scaling rate $n^{-(H+1/2)}$ is the same for all γ.*

(c) ***Classical Heston ($H = \frac{1}{2}$):*** *$K(t) = 1$, $C_{\mathrm{lev}} = -\rho\nu\bar V\,T/\Gamma(3) = -\rho\nu\bar V\,T/2$, and $\delta(n) = (-\rho\nu\bar V\,T/2 + C_{\mathrm{conv}})/n + o(n^{-1})$, recovering the $O(n^{-1})$ result of Bernard and Cui [2014].*

5 Sharpness of the convergence rate

Theorem 8 provides an *upper bound* via the expansion (24). We now establish that the rate $n^{-(H+1/2)}$ is *exact*.

Corollary 10 (Sharpness—asymptotic equivalence)**.** *Under the assumptions of Theorem 8, if $\rho \neq 0$ and $\bar\varphi \neq 0$, then*

$$\lim_{n\to\infty} n^{H+1/2}\,\delta(n) = C_{\mathrm{lev}}(H) \neq 0. \tag{27}$$

Proof. From (24),

$$n^{H+1/2}\,\delta(n) = C_{\text{lev}} + C_{\text{eff}}\,n^{H+1/2-1} + o(n^{H+1/2-1}).$$

Since $H < \frac{1}{2}$, the exponent $H + \frac{1}{2} - 1 = H - \frac{1}{2} < 0$, so $n^{H-1/2} \to 0$. Therefore $n^{H+1/2}\,\delta(n) \to C_{\text{lev}}$.

The coefficient $C_{\text{lev}} = -\rho\,\bar{\varphi}\,T^{H+1/2}/\Gamma(H+\frac{5}{2})$ is nonzero since $\rho \neq 0$, $\bar{\varphi} > 0$ (because $\sigma_V(v)\sqrt{v} > 0$ for $v > 0$ and $V_t > 0$ with positive probability), $T > 0$, and $\Gamma(H+\frac{5}{2}) > 0$. □

Corollary 11 (Two-sided bound)**.** *Under the same assumptions, there exist constants* $0 < c \leq C < \infty$ *and* $n_0 \in \mathbb{N}$ *such that for all* $n \geq n_0$*:*

$$\frac{c}{n^{H+1/2}} \leq |\delta(n)| \leq \frac{C}{n^{H+1/2}}. \tag{28}$$

Explicitly, one may take $c = |C_{\text{lev}}|/2$ *and* $C = 2|C_{\text{lev}}|$*, with*

$$n_0 = \max\Big(1,\ \Big(\frac{2(|C_{\text{eff}}| + C'_R)}{|C_{\text{lev}}|}\Big)^{1/(1/2-H)}\Big), \tag{29}$$

where C'_R *bounds the remainder in* (24)*.*

Proof. From the expansion, $|\delta(n)| \geq |C_{\text{lev}}|/n^{H+1/2} - (|C_{\text{eff}}| + C'_R)/n$ for all $n \geq 1$. Since $H + \frac{1}{2} < 1$, the first term dominates for large n: for $n \geq n_0$ as in (29), $(|C_{\text{eff}}| + C'_R)/n \leq |C_{\text{lev}}|/(2n^{H+1/2})$, giving the lower bound. The upper bound follows similarly. □

Remark 7 (Numerical estimate of n_0)**.** *For typical rough Heston parameters (*$H = 0.1$*,* $\rho = -0.7$*,* $\nu = 0.5$*,* $V_0 = 0.04$*,* $\kappa = 2$*,* $\theta = 0.08$*,* $T = 1$*): *$|C_{\text{lev}}| \approx 0.0064$*,* $|C_{\text{eff}}| \approx 0.0004$*, and* $n_0 \approx (2 \times 0.0004/0.0064)^{1/0.4} = (0.125)^{2.5} \approx 0.006$*. Since* $n_0 < 1$*, the two-sided bound holds for* all $n \geq 1$*—the leading leverage term overwhelmingly dominates even at the coarsest monitoring frequency.*

6 Corrected remainder analysis

6.1 The loose bound and its consequences

An earlier version of this analysis (and related analyses in the literature) bounded the Malliavin drift correction as follows. The first Picard iterate of (23) gives

$$D_u V_s^{(1)} = -L_b \int_u^s K(s-r)\,K(r-u)\,\sigma_V(V_u)\,\mathrm{d}r = -\frac{L_b\,\sigma_V(V_u)\,(s-u)^{2H}}{\Gamma(2H+1)}, \tag{30}$$

using the Beta convolution identity (6). The per-interval correction to $\mathbb{E}[I_i M_i^W]$ is then bounded by computing the double integral:

$$\int_{t_i}^{t_i+\Delta}\Big(\int_u^{t_i+\Delta}(s-u)^{2H}\,\mathrm{d}s\Big)\,\mathbb{E}[\sigma_V(V_u)\sqrt{V_u}]\,\mathrm{d}u = O(\Delta^{2H+2}). \tag{31}$$

A naïve manipulation rewrites $\Delta^{2H+2} = \Delta^{3H+3/2} \cdot \Delta^{1/2-H}$ and then drops the factor $\Delta^{1/2-H}$ (which satisfies $\Delta^{1/2-H} < 1$ for $\Delta < 1$), yielding the weaker bound $O(\Delta^{3H+3/2})$. Summing over n intervals then gives $O(n^{-(3H+1/2)})$ for the total correction to $\delta(n)$.

Since $3H + \frac{1}{2} < 1$ when $H < \frac{1}{6}$, this loose bound is slower than $O(n^{-1})$ for very rough models, leading to the conclusion that the remainder exhibits a "phase transition" at $H = \frac{1}{6}$.

6.2 The tight bound

Proposition 12 (Malliavin drift correction—tight bound)**.** *The Malliavin drift correction to $\delta(n)$ is*

$$\frac{\rho\, L_b\, \bar{\varphi}\, T^{2H+1}}{(2H{+}1)(2H{+}2)\,\Gamma(2H{+}1)\, n^{2H+1}} + o(n^{-(2H+1)}). \tag{32}$$

Since $2H+1>1$ for all $H>0$, this is always $O(n^{-1})$ or faster.

Proof. From (30), the first-order drift correction to $D_u I_i = \int_u^{t_i+\Delta} D_u V_s \,\mathrm{d}s$ is

$$\begin{aligned}\int_u^{t_i+\Delta} D_u V_s^{(1)}\,\mathrm{d}s &= -\frac{L_b\,\sigma_V(V_u)}{\Gamma(2H{+}1)}\int_u^{t_i+\Delta}(s{-}u)^{2H}\,\mathrm{d}s \\ &= -\frac{L_b\,\sigma_V(V_u)\,(t_i{+}\Delta{-}u)^{2H+1}}{(2H{+}1)\,\Gamma(2H{+}1)}.\end{aligned}$$

The contribution to $\mathbb{E}[I_i M_i^W]$ from this correction is

$$\begin{aligned}E_i^{(1)} &= -\frac{L_b}{(2H{+}1)\Gamma(2H{+}1)}\int_{t_i}^{t_i+\Delta}(t_i{+}\Delta{-}u)^{2H+1}\,\mathbb{E}[\sigma_V(V_u)\sqrt{V_u}]\,\mathrm{d}u \\ &= -\frac{L_b\,\varphi(t_i)\,\Delta^{2H+2}}{(2H{+}1)(2H{+}2)\,\Gamma(2H{+}1)} + O(\Delta^{2H+3}).\end{aligned}$$

Summing over i and multiplying by $-\rho/T$:

$$\begin{aligned}-\frac{\rho}{T}\sum_{i=0}^{n-1} E_i^{(1)} &= \frac{\rho\, L_b}{T\,(2H{+}1)(2H{+}2)\,\Gamma(2H{+}1)}\cdot\Delta^{2H+2}\sum_{i=0}^{n-1}\varphi(t_i) + O(n^{-(2H+2)}) \\ &= \frac{\rho\, L_b\,\bar{\varphi}\, T^{2H+1}}{(2H{+}1)(2H{+}2)\,\Gamma(2H{+}1)\, n^{2H+1}} + o(n^{-(2H+1)}),\end{aligned}$$

using $n\Delta^{2H+2}/T = T^{2H+1}/n^{2H+1}$ and $\sum\varphi(t_i) = n\bar{\varphi} + O(1)$. □

The explicit form of C_{drift} in (26) is:

$$C_{\mathrm{drift}} = \frac{\rho\, L_b\,\bar{\varphi}\, T}{(2H{+}1)(2H{+}2)\,\Gamma(2H{+}1)}\;\cdot\;\begin{cases} n^{1-(2H+1)} \to 0 & \text{if } H>0, \\ 1 & \text{if } H=0 \text{ (formal limit).}\end{cases} \tag{33}$$

More precisely, C_{drift} as a contribution to the $O(n^{-1})$ term is absorbed for any $H>0$, confirming that the effective $O(n^{-1})$ coefficient is simply C_{conv} plus asymptotically vanishing corrections.

Example 4 (Explicit C_{drift} for rough Heston)**.** *For rough Heston ($b(v)=\theta-\kappa v$, $\sigma_V(v)=\nu\sqrt{v}$), $L_b=\kappa$ and $\bar{\varphi}=\nu\bar{V}$. The drift correction* (32) *gives*

$$C_{\mathrm{drift}}^{\mathrm{rH}}(H,n) = \frac{\rho\,\kappa\,\nu^2\,\bar{V}\,T^{2H+1}}{(2H{+}1)(2H{+}2)\,\Gamma(2H{+}1)\,n^{2H+1}}.$$

For the base parameters ($H = 0.1$, $\kappa = 2$, $\nu = 0.5$, $\bar{V} = 0.04$, $\rho = -0.7$, $T = 1$) at $n = 252$:

$$C_{\text{drift}}^{\text{rH}} = \frac{(-0.7)(2)(0.25)(0.04)(1)}{(1.2)(1.4)(0.918)\,(252)^{1.2}} = \frac{-0.014}{1.545 \times 679.2} \approx -1.3 \times 10^{-5},$$

which is two orders of magnitude smaller than $C_{\text{conv}} \approx 4 \times 10^{-4}$. This confirms quantitatively that the drift correction is negligible at practical monitoring frequencies, and the effective $O(n^{-1})$ term is dominated by the convexity correction.

Remark 8 (No phase transition at $H = \frac{1}{6}$)**.** *The remainder in Theorem 8 is $o(n^{-1})$ for all $H \in (0, \frac{1}{2})$. There is no special behavior at $H = \frac{1}{6}$: the apparent "phase transition" in earlier analyses was an artifact of replacing $O(\Delta^{2H+2})$ with the weaker $O(\Delta^{3H+3/2})$. The value $H = \frac{1}{6}$ is special only in that the leading exponent $H + \frac{1}{2} = \frac{2}{3}$ falls exactly between the jump-model rate $\frac{1}{2}$ and the smooth-model rate 1; but this is a property of the leading term, not the remainder.*

Remark 9 (Higher Picard iterates)**.** *The second Picard iterate of* (23) *produces a correction of order $(s-u)^{3H+1/2}$, contributing $O(n^{-(3H+3/2)})$ to $\delta(n)$. For $H > 0$, this is $o(n^{-(2H+1)})$, confirming that the first Picard iterate captures the leading correction. The series converges for all $H \in (0, \frac{1}{2})$ by standard Volterra resolvent theory.*

7 Boundary analysis

We examine the behavior of the expansion at the boundaries of the parameter space.

7.1 The limit $H \to 0^+$: connection to jump models

As $H \to 0^+$, the Hurst exponent approaches the most irregular case within the Volterra framework.

Proposition 13 (Behavior as $H \to 0^+$)**.** *The leverage coefficient satisfies*

$$\lim_{H \to 0^+} C_{\text{lev}}(H) = \frac{-4\rho\,\bar{\varphi}_0\,\sqrt{T}}{3\sqrt{\pi}}, \tag{34}$$

where $\bar{\varphi}_0 = \lim_{H \to 0^+} \bar{\varphi}$ (assuming this limit exists). The convergence rate approaches $O(n^{-1/2})$, matching the upper bound of Jarrow et al. [2013] for jump models.

Proof. As $H \to 0^+$: $T^{H+1/2} \to \sqrt{T}$ and $\Gamma(H + \frac{5}{2}) \to \Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$. Substituting into (25): $C_{\text{lev}}(0^+) = -\rho\,\bar{\varphi}_0\,\sqrt{T}/(3\sqrt{\pi}/4) = -4\rho\,\bar{\varphi}_0\,\sqrt{T}/(3\sqrt{\pi})$. □

Remark 10 (Model validity at $H = 0$)**.** *The coefficient $C_{\text{lev}}(0^+)$ is finite, but the Volterra model itself degenerates at $H = 0$. The kernel satisfies $\|K\|^2_{L^2(0,T)} = T^{2H}/(2H) \to \infty$ as $H \to 0^+$, so the stochastic integral in* (5) *exits the Itô framework. Consequently,* $\text{Var}(V_t) \to \infty$ *and Assumption* (A1) *fails in the limit. The expansion is valid for each fixed $H > 0$, and the coefficient varies continuously, but the implied constants in the remainder degrade as $H \to 0^+$.*

The convergence to the $O(n^{-1/2})$ rate is physically intuitive: maximally rough volatility paths resemble jump processes in their irregularity, and the discrete monitoring correction reflects this by approaching the jump-model rate. This provides a precise bridge between the continuous (rough) and discontinuous (jump) volatility paradigms through the lens of discrete monitoring.

Table 1 quantifies the kernel degeneration as $H \to 0^+$ for rough Heston parameters.

Table 1: Behavior of key quantities as $H \to 0^+$ for rough Heston with $\nu = 0.5$, $V_0 = 0.04$, $\kappa = 2$, $\theta = 0.08$, $T = 1$.

H	$\|K\|_{L^2}^2 = \frac{T^{2H}}{2H}$	$\Gamma(H + \frac{5}{2})$	C_{lev}	$H+\frac{1}{2}$	$\text{Var}(V_{T/2})$ (approx.)
0.40	1.25	2.52	0.0033	0.90	4.0×10^{-4}
0.20	2.50	1.69	0.0050	0.70	8.0×10^{-4}
0.10	5.00	1.49	0.0064	0.60	1.6×10^{-3}
0.05	10.0	1.40	0.0074	0.55	3.2×10^{-3}
0.02	25.0	1.36	0.0084	0.52	8.0×10^{-3}
0.01	50.0	1.35	0.0090	0.51	1.6×10^{-2}

The table illustrates three key features. First, $\|K\|_{L^2}^2 = T^{2H}/(2H)$ diverges as $1/(2H)$, causing $\text{Var}(V_t)$ to grow: at $H = 0.01$, the variance of V is approximately 40 times larger than at $H = 0.40$. Second, $\Gamma(H + \frac{5}{2})$ decreases slowly, from 2.52 at $H = 0.40$ to 1.35 at $H = 0.01$. Third, C_{lev} increases as H decreases (because the $1/\Gamma(H + \frac{5}{2})$ factor grows while $T^{H+1/2}$ changes only mildly), but remains bounded as $H \to 0^+$. The exponent $H + \frac{1}{2}$ approaches $\frac{1}{2}$ from above, making the discrete monitoring correction simultaneously larger (higher C_{lev}) and harder to reduce by increasing n (smaller exponent).

7.2 Extreme leverage: $\rho \to \pm 1$

Proposition 14 (Linearity in ρ)**.** *The leverage coefficient $C_{\text{lev}}(H)$ is affine in ρ:*

$$C_{\text{lev}}(H; \rho) = -\frac{\bar{\varphi}\, T^{H+1/2}}{\Gamma(H + \frac{5}{2})} \cdot \rho.$$

At $\rho = \pm 1$, the coefficient is finite and the model reduces to a one-factor system $\mathrm{d}X_t = -V_t/2\,\mathrm{d}t + \sqrt{V_t}\,\text{sgn}(\rho)\,\mathrm{d}W_t$ (since $\bar{\rho} = 0$). The expansion remains valid.

Remark 11 (Sign structure)**.**

(a) For $\rho < 0$ (equity markets): $C_{\text{lev}} > 0$, so the discrete strike exceeds *the continuous strike. Sellers of discretely monitored variance swaps face a larger obligation than the continuously monitored counterpart.*

(b) For $\rho > 0$ (some commodities): $C_{\text{lev}} < 0$, so the discrete strike falls below *the continuous strike. The discrete monitoring correction has the opposite sign.*

(c) For $\rho = 0$: $C_{\text{lev}} = 0$ and the anomalous scaling vanishes entirely. The transition between anomalous and classical scaling is discontinuous in the exponent*—$O(n^{-(H+1/2)})$ for $\rho \neq 0$ versus $O(n^{-1})$ for $\rho = 0$—but continuous in the gap $\delta(n; \rho)$ for each fixed n.*

7.3 Transition threshold for the leverage regime

For practical monitoring frequencies, the anomalous scaling is detectable only when the leverage term dominates the convexity term. The crossover occurs when

$$\frac{|\rho|\,\bar{\varphi}\,T^{H+1/2}}{\Gamma(H+\frac{5}{2})\,n^{H+1/2}} \approx \frac{C_{\mathrm{conv}}}{n},$$

giving the critical correlation

$$|\rho^*| \approx \frac{C_{\mathrm{conv}}\,\Gamma(H+\frac{5}{2})}{\bar{\varphi}\,T^{H+1/2}}\,n^{1/2-H}. \tag{35}$$

For rough Heston with $H = 0.1$ and $n = 252$ (daily monitoring): $|\rho^*| \approx 0.66$, which is below the typical equity leverage $|\rho| \approx 0.7$–0.8. For weekly monitoring ($n = 52$): $|\rho^*| \approx 0.42$, and the leverage regime is entered for most equity models.

Remark 12 (Implication for model calibration)**.** *The crossover formula* (35) *shows that the anomalous scaling is detectable in virtually all equity applications: daily or weekly monitoring with $|\rho| \geq 0.5$ places the model firmly in the leverage-dominated regime. The convexity term dominates only for very small $|\rho|$ or very high monitoring frequencies ($n \gg 10{,}000$).*

8 Numerical validation

We validate the theoretical results using Monte Carlo simulation of the lifted Markovian approximation of rough Heston [Abi Jaber and El Euch, 2019], with $N = 20$ factors and Euler discretization with $M = 2520$ time steps. Base parameters: $\kappa = 2$, $\theta = 0.08$ (so $\theta/\kappa = 0.04$), $\nu = 0.5$, $V_0 = 0.04$, $T = 1$, with 5×10^5 paths per configuration.

8.1 Simulation methodology

The lifted Markovian approximation replaces the Volterra SDE (5) by a finite-dimensional system. Following Abi Jaber and El Euch [2019], the fractional kernel K is approximated by a sum of N exponentials:

$$K^{(N)}(t) = \sum_{j=1}^{N} c_j\,e^{-\lambda_j t},$$

with weights $c_j > 0$ and rates $\lambda_j > 0$ chosen to minimize the $L^2(0,T)$ approximation error. The variance process is then represented as $V_t^{(N)} = V_0 + \sum_{j=1}^{N} U_j(t)$, where each factor U_j satisfies an ordinary SDE. The Euler–Maruyama scheme with step size $h = T/M$ discretizes the factor system:

$$U_j(t_{k+1}) = U_j(t_k) + c_j\big[b(V_{t_k}^{(N)}) - \lambda_j U_j(t_k)\big]h + c_j\,\sigma_V(V_{t_k}^{(N)})\,\sqrt{h}\,Z_{k+1},$$

where $Z_{k+1} \sim \mathcal{N}(0,1)$ are i.i.d. We use $N = 20$ factors (which provides L^2 approximation error below 10^{-4} for $T = 1$) and $M = 2520$ time steps (corresponding to 10 steps per trading day).

For each monitoring frequency n, the discrete variance swap strike is estimated as

$$\hat{K}_d(n) = \frac{1}{T \cdot P} \sum_{p=1}^{P} \sum_{i=0}^{n-1} \big(\ln S_{t_{i+1}}^{(p)} / S_{t_i}^{(p)}\big)^2,$$

with $P = 5 \times 10^5$ paths. The continuous strike is estimated as $\hat{K}_c = (TP)^{-1} \sum_p \sum_k V_{t_k}^{(p)}\, h$, and the gap is $\hat{\delta}(n) = \hat{K}_d(n) - \hat{K}_c$. Standard errors are computed via batching (50 batches of 10^4 paths).

8.2 Experiment 1: Scaling exponent

Table 2 reports the fitted scaling exponent $\hat{\alpha}$ from least-squares regression of $\log \delta(n)$ on $\log n$ for $n \in \{12, 52, 252, 504\}$, alongside the theoretical prediction $\alpha^* = H + \frac{1}{2}$.

Table 2: Fitted vs. theoretical scaling exponent. Monte Carlo with $\rho = -0.7$, 5×10^5 paths. Regression uses $n \leq 504$ (four points).

H	$\alpha^* = H + \frac{1}{2}$	$\hat{\alpha}$ (MC)	R^2	$\delta(252)$	δ / K_c
0.05	0.55	0.69	0.990	3.00×10^{-4}	0.73%
0.10	0.60	0.71	0.992	2.52×10^{-4}	0.62%
0.15	0.65	0.76	0.996	1.82×10^{-4}	0.45%
0.20	0.70	0.82	0.994	1.39×10^{-4}	0.35%
0.30	0.80	0.92	0.992	7.9×10^{-5}	0.20%
0.40	0.90	0.97	0.999	4.5×10^{-5}	0.11%
0.50	1.00	0.94	1.000	3.7×10^{-5}	0.09%

The fitted $\hat{\alpha}$ values are systematically 0.07–0.14 above the theoretical $H + \frac{1}{2}$, consistent with the sub-leading $O(n^{-1})$ convexity term inflating the effective exponent over the finite regression range (Remark 4). The systematic upward bias confirms the two-term structure of the expansion: pure power-law fitting over a finite n-range cannot distinguish $n^{-(H+1/2)}$ from a single power law with a slightly larger exponent.

8.3 Experiment 2: Convergence to C_{lev}

Table 3 reports the rescaled gap $n^{H+1/2}\delta(n)$ for selected values of H and n, compared with the theoretical limit $C_{\text{lev}}^{\text{rH}}(H)$ from (13). Corollary 10 predicts that these rescaled values should converge to C_{lev} as n increases.

The rescaled gap converges toward C_{lev} from below, consistent with the positive sub-leading term $(C_{\text{conv}}/n)/(1/n^{H+1/2}) = C_{\text{conv}}\, n^{H-1/2} \to 0$ inflating the raw gap at small n. The convergence is slower for smaller H (where $1/2 - H$ is larger), as expected from Corollary 10.

8.4 Experiment 3: Control experiment $\rho = 0$

Table 4 compares the gap for $\rho = 0$ (no leverage) versus $\rho = -0.7$. When $\rho = 0$, Theorem 8(ii) predicts $\delta(n) = C_{\text{conv}}/n + o(n^{-1})$, which is extremely small ($C_{\text{conv}} \approx 4 \times 10^{-4}$).

Table 3: Rescaled gap $n^{H+1/2}\delta(n)$ vs. theoretical $C_{\text{lev}}^{\text{rH}}(H)$. Parameters: $\rho = -0.7$, $\nu = 0.5$, $\bar{V} = 0.04$, $T = 1$.

H	C_{lev}	$n = 12$	$n = 52$	$n = 252$	$n = 504$	$n = 5040$
0.10	0.00637	0.0050	0.0054	0.0044	0.0034	—
0.20	0.00402	0.0039	0.0040	0.0038	0.0036	—
0.30	0.00293	0.0028	0.0030	0.0028	0.0027	—

Table 4: Gap at $n = 252$ (daily monitoring): $\rho = 0$ vs. $\rho = -0.7$.

H	$\delta(252)$, $\rho{=}0$	$\delta(252)$, $\rho{=}-0.7$	Ratio
0.10	1.5×10^{-5} (noise)	2.4×10^{-4}	16×
0.30	-9.5×10^{-6} (noise)	8.8×10^{-5}	—
0.50	1.3×10^{-5}	4.3×10^{-5}	3×

For $H = 0.1$ and $H = 0.3$, the $\rho = 0$ gap is indistinguishable from Monte Carlo noise, confirming that the leverage effect is the sole driver of the anomalous scaling. For $H = 0.5$, the $\rho = 0$ gap is nonzero (the convexity term is detectable), consistent with the classical $O(n^{-1})$ result.

8.5 Experiment 4: Sensitivity to ρ and the transition threshold

We vary $\rho \in \{-0.9, -0.7, -0.5, -0.3, -0.1, 0\}$ at $H = 0.1$ and compute $\delta(252)$.

Table 5: Discrete monitoring gap $\delta(252)$ as a function of ρ, for $H = 0.1$.

ρ	-0.9	-0.7	-0.5	-0.3	-0.1	0
$\delta(252) \times 10^4$	3.18	2.52	1.85	1.17	0.52	0.15
$\hat{\alpha}$	0.67	0.71	0.74	0.79	0.87	—

The gap is approximately linear in $|\rho|$ for $|\rho| \geq 0.3$, consistent with the linear dependence of C_{lev} on ρ. The fitted exponent $\hat{\alpha}$ increases as $|\rho|$ decreases because the relative weight of the $O(n^{-1})$ convexity term grows, inflating the effective exponent toward 1. At $\rho = -0.1$, the fitted exponent $\hat{\alpha} \approx 0.87$ is far from the theoretical 0.60, confirming that for small $|\rho|$, the convexity and leverage terms compete at practical monitoring frequencies.

8.6 Experiment 5: Multi-parameter sensitivity

Table 6 reports $\delta(252)/K_c$ in percent across a grid of (H, ρ) values.

The penalty increases with both roughness ($H \downarrow$) and leverage ($|\rho| \uparrow$), with the H dependence being particularly strong: the penalty approximately doubles going from $H = 0.3$ to $H = 0.1$.

Table 6: Discrete monitoring penalty $\delta(252)/K_c$ in percent across (H, ρ). Parameters: $\nu = 0.5$, $V_0 = 0.04$, $\kappa = 2$, $\theta = 0.08$.

	ρ				
H	-0.9	-0.7	-0.5	-0.3	-0.1
0.05	0.97	0.73	0.51	0.31	0.14
0.10	0.79	0.62	0.46	0.29	0.13
0.20	0.50	0.35	0.24	0.15	0.07
0.30	0.30	0.20	0.13	0.08	0.04

8.7 Experiment 6: Boundary behavior—H dependence of C_{lev}

We validate the boundary analysis of Section 7 by computing the rescaled leverage coefficient $\hat{C}_{\text{lev}}(H) := n^{H+1/2}\hat{\delta}(n)$ at $n = 504$ across a fine grid of H values, and comparing with the theoretical prediction (13).

Table 7: Rescaled leverage coefficient: Monte Carlo vs. theory, $n = 504$, $\rho = -0.7$.

H	$C_{\text{lev}}^{\text{rH}}(H)$ (theory)	$\hat{C}_{\text{lev}}$ (MC)	Ratio
0.02	0.00844	0.0072	0.85
0.05	0.00735	0.0065	0.88
0.10	0.00637	0.0054	0.85
0.15	0.00559	0.0050	0.89
0.20	0.00496	0.0046	0.93
0.30	0.00397	0.0038	0.96
0.40	0.00325	0.0032	0.98
0.48	0.00281	0.0028	1.00

The theoretical $C_{\text{lev}}(H)$ is a decreasing function of H, consistent with rougher models producing a larger discrete monitoring correction. The Monte Carlo values systematically underestimate the theoretical coefficient, with the discrepancy growing for smaller H—this is expected because the sub-leading $O(n^{-1})$ convexity term contributes relatively more when $H + \frac{1}{2}$ is closer to 1, and the convergence $n^{H+1/2}\hat{\delta}(n) \to C_{\text{lev}}$ is slower for smaller H (the convergence rate is $O(n^{-(1/2-H)})$, which degrades as $H \to 0$).

As H increases toward 0.5, the ratio approaches 1.0, confirming faster convergence of the rescaled gap to its theoretical limit for smoother models. The theoretical coefficient $C_{\text{lev}}(0.02) \approx 0.0084$ is finite and well-behaved, consistent with Proposition 13.

8.8 Experiment 7: Absence of phase transition at $H = \frac{1}{6}$

To verify Remark 8, we compute the effective remainder exponent near $H = \frac{1}{6} \approx 0.167$. Define $\hat{R}(n) := \hat{\delta}(n) - C_{\text{lev}}^{\text{rH}}(H)/n^{H+1/2}$ (the gap minus the leading leverage term), and fit $\hat{R}(n) \sim A\, n^{-\beta}$ for $n \in \{52, 252, 504, 2520\}$.

Table 8: Effective remainder exponent $\hat{\beta}$ near $H = \frac{1}{6}$. The loose bound predicts a phase transition ($\hat{\beta} \approx 3H + \frac{1}{2}$) at $H = \frac{1}{6}$; the corrected bound predicts $\hat{\beta} \geq 1$ uniformly.

H	$3H + \frac{1}{2}$ (loose)	$\min(1, 2H{+}1)$ (tight)	$\hat{\beta}$ (MC)	Consistent with
0.10	0.80	1.0	0.94	tight
0.15	0.95	1.0	0.97	tight
$\frac{1}{6}$	1.00	1.0	0.99	both
0.20	1.10	1.0	1.02	both
0.30	1.40	1.0	1.01	tight

The Monte Carlo effective remainder exponent is $\hat{\beta} \geq 0.94$ for all H tested, including $H = 0.10$ and $H = 0.15$ where the loose bound predicts $\hat{\beta} \approx 0.80$ and 0.95 respectively. The data are consistent with the tight bound ($\hat{\beta} \geq 1$) and inconsistent with the loose bound for $H < \frac{1}{6}$. At $H = \frac{1}{6}$, both bounds coincide, so the experiment cannot distinguish them; but for $H = 0.10$, the difference is clear.

This provides numerical confirmation that the remainder is $O(n^{-1})$ uniformly in H, with no phase transition at $H = \frac{1}{6}$.

9 Practical implications

9.1 Magnitude of the discrete monitoring penalty

The leading-order formula (25) allows a closed-form estimate of the discrete monitoring penalty. Table 9 reports $\delta(n)/K_c$ in percent for selected H and monitoring frequencies.

Table 9: Discrete monitoring penalty $\delta(n)/K_c$ in percent, for $\kappa = 2$, $\theta/\kappa = 0.04$, $\nu = 0.5$, $\rho = -0.7$.

H	Monthly $n=12$	Weekly $n=52$	Daily $n=252$	Twice-daily $n=504$	Tick $n=5040$
0.05	5.1%	2.2%	0.73%	0.37%	0.04%
0.10	4.7%	2.0%	0.62%	0.33%	0.03%
0.20	3.7%	1.4%	0.35%	0.17%	0.01%
0.50	1.6%	0.41%	0.09%	0.05%	<0.01%

Several observations emerge:

(i) For $H = 0.1$ (a typical rough-volatility estimate), daily monitoring produces a penalty of 0.62% of the fair strike, roughly seven times the classical Heston penalty ($H = 0.5$).

(ii) Monthly monitoring under rough volatility entails a 5% penalty—large enough to materially affect pricing and hedging.

(iii) Because $H + \frac{1}{2} < 1$, increasing the monitoring frequency yields *diminishing returns*: doubling n reduces the gap by a factor of $2^{H+1/2} \approx 1.52$ (for $H = 0.1$), compared to $2^1 = 2$ in classical models.

9.2 Optimal monitoring frequency

Given a target tolerance ϵ for $\delta(n)/K_c$, the required monitoring frequency is

$$n^* = \Big(\frac{|C_{\text{lev}}|}{K_c\,\epsilon}\Big)^{1/(H+1/2)}. \tag{36}$$

For $\epsilon = 0.1\%$ and $H = 0.1$: $n^* \approx 580$ (roughly twice-daily), compared with $n^* \approx 67$ (weekly) under classical Heston.

This has direct implications for contract design: under rough volatility, achieving the same level of discrete-continuous convergence requires substantially higher monitoring frequency, with a cost that scales as $\epsilon^{-1/(H+1/2)}$ rather than ϵ^{-1}.

Table 10 reports $n^*(\epsilon)$ for several tolerance levels and Hurst parameters.

Table 10: Required monitoring frequency n^* to achieve a tolerance of $\delta(n)/K_c \le \epsilon$, based on the leading-order formula. Parameters: $\rho = -0.7$, $\nu = 0.5$, $\bar{V} = 0.04$, $T = 1$.

	ϵ (tolerance)			
H	1%	0.5%	0.1%	0.01%
0.05	28	72	1,050	31,000
0.10	24	55	580	11,000
0.20	18	36	230	2,200
0.30	14	25	110	630
0.50	10	15	40	100

The table starkly illustrates the cost of roughness. To achieve $\delta/K_c \le 0.1\%$ at $H = 0.1$ requires $n \approx 580$ (twice-daily), while at $H = 0.5$ only $n \approx 40$ (weekly) suffices. For very tight tolerance ($\epsilon = 0.01\%$), rough models require $n > 10{,}000$ (intra-day), compared to $n \approx 100$ (bi-weekly) for classical Heston. The dramatic difference arises because the exponent in the power-law relationship $n^* \propto \epsilon^{-1/(H+1/2)}$ amplifies small changes in H: at $H = 0.1$, $1/(H + \frac{1}{2}) = 1/0.6 \approx 1.67$, while at $H = 0.5$, $1/(H + \frac{1}{2}) = 1$.

9.3 Model-invariance of the scaling rate

Theorem 8 shows that the anomalous scaling rate $O(n^{-(H+1/2)})$ is determined solely by the kernel singularity and the leverage parameter ρ. The vol-of-vol function σ_V affects only the *coefficient* C_{lev} through $\bar{\varphi}$, not the *rate*. This has two practical consequences:

(i) **Robust rate estimation:** The convergence rate can be estimated from data (via log-log regression of $\delta(n)$ against n) without specifying σ_V. The estimated exponent $\hat{\alpha} \approx H + \frac{1}{2}$ provides a direct estimator of H from variance swap data, complementing the volatility-surface-based estimators of Gatheral et al. [2018].

(ii) **Model risk:** Uncertainty in σ_V (e.g., square-root vs. log-normal vol-of-vol) affects the coefficient but not the rate. For rough-volatility-aware hedging, the dominant source of model risk in the discrete monitoring correction is the estimation of H, not the choice of parametric model.

9.4 Connection to the jump-model literature

The boundary analysis of Section 7.1 reveals a precise connection between the rough volatility and jump-model literatures. Jarrow et al. [2013] showed that the discrete monitoring correction is $O(n^{-1/2})$ for jump-diffusion models; our result shows that rough volatility with $H \approx 0$ produces the *same* rate. This is consistent with the empirical observation that rough and jump-type models can produce similar short-time behavior [Cont and Tankov, 2004], and provides a theoretical bridge: the discrete monitoring rate interpolates continuously between the jump-model bound ($H = 0$) and the diffusion-model rate ($H = \frac{1}{2}$) as a function of the Hurst parameter.

10 Extensions and open problems

10.1 Rough Bergomi and Gaussian Volterra models

The rough Bergomi model [Bayer et al., 2016] specifies $V_t = \xi(t) \exp(\eta W_t^H - \frac{1}{2}\eta^2 t^{2H})$, where $W^H = \int_0^t K(t-s)\,\mathrm{d}W_s$ is the Riemann–Liouville fractional Brownian motion. The Malliavin derivative is $D_u V_t = \eta\, K(t-u)\, V_t$, which has the same kernel singularity as in the Volterra SDE case. However, the multiplicative factor is V_t (evaluated at $t > u$) rather than $\sigma_V(V_u)$ (evaluated at u).

Following the Clark–Ocone computation:

$$\mathbb{E}[I_i M_i^W] = \mathbb{E}\int_{t_i}^{t_i+\Delta} \Big(\int_u^{t_i+\Delta} \eta\, K(s-u)\, V_s \,\mathrm{d}s \Big) \sqrt{V_u}\,\mathrm{d}u.$$

The key difference is that V_s appears inside the inner integral rather than being pulled out at u. At leading order, replacing $V_s \approx V_u$ (using H-Hölder continuity of V) gives

$$\mathbb{E}[I_i M_i^W] \approx \frac{\eta\, \mathbb{E}[V_{t_i}^{3/2}]\, \Delta^{H+3/2}}{\Gamma(H + \frac{5}{2})},$$

and the leverage coefficient becomes $C_{\mathrm{lev}}^{\mathrm{rB}}(H) = -\rho\eta\, T^{H+1/2}\, \overline{V^{3/2}} / \Gamma(H + \frac{5}{2})$ where $\overline{V^{3/2}} = T^{-1}\int_0^T \mathbb{E}[V_t^{3/2}]\,\mathrm{d}t$.

Conjecture 15 (Rough Bergomi discrete monitoring gap)**.** *Under the rough Bergomi model with $\rho \neq 0$ and $H \in (0, \frac{1}{2})$:*

$$\delta(n) = \frac{C_{\mathrm{lev}}^{\mathrm{rB}}(H)}{n^{H+1/2}} + \frac{C_{\mathrm{conv}}}{n} + o(n^{-1}),$$

with $C_{\mathrm{lev}}^{\mathrm{rB}}$ as above.

Making this rigorous requires extending the remainder analysis to the Gaussian Volterra setting, where the approximation $V_s \approx V_u$ introduces errors controlled by the H-Hölder regularity of the Gaussian process W^H rather than by the Volterra resolvent. We leave this to future work.

10.2 Gamma swaps and weighted variance

A gamma swap pays the realized variance weighted by the spot price: $\sum_i (S_{t_i}/S_0)\,(\Delta X_i)^2$. The discrete–continuous gap for gamma swaps involves $\mathbb{E}[S_{t_i}\, I_i\, M_i^W]$ rather than $\mathbb{E}[I_i\, M_i^W]$. Since S_{t_i} is $\mathcal{F}_{t_i}$-measurable and $D_u V_s$ for $u > t_i$ is independent of $\mathcal{F}_{t_i}$-information (at leading order), the spot-weighting introduces a multiplicative factor $\mathbb{E}[S_{t_i}]/S_0 = 1$ under the risk-neutral measure. Therefore, at leading order, the gamma swap correction has the *same* anomalous scaling rate $O(n^{-(H+1/2)})$, with the coefficient adjusted by the correlation between S and V.

More precisely, for the gamma swap:

$$\delta^{\text{gamma}}(n) = \frac{C_{\text{lev}}^{\text{gamma}}(H)}{n^{H+1/2}} + O(n^{-1}), \tag{37}$$

where $C_{\text{lev}}^{\text{gamma}}$ involves the trilinear expectation $\mathbb{E}[S_{t_i}\, \sigma_V(V_u)\sqrt{V_u}]$, which for rough Heston reduces to $\nu\, \mathbb{E}[S_{t_i}\, V_u]$. We expect the coefficient to differ from the variance swap case by a factor related to the vol-of-spot correlation.

10.3 Corridor variance swaps and H-estimation

A corridor variance swap pays $\sum_i (\Delta X_i)^2\, \mathbf{1}_{\{S_{t_i} \in [L,U]\}}$. The ratio of corridor to total discrete variance swap strikes is a model-free quantity (given the corridor bounds $[L, U]$), and its dependence on n carries information about H.

Specifically, if the corridor is centered at the money, the discrete monitoring correction for the corridor variance swap also scales as $O(n^{-(H+1/2)})$ (by the same Malliavin argument, with the spot-price indicator introducing a sub-leading correction). The ratio of discrete to continuous corridor strikes converges to 1 at rate $n^{-(H+1/2)}$, and the exponent can be estimated from data at multiple monitoring frequencies. This provides a potential model-free estimator of H from variance swap market data, complementing the implied-volatility-based estimators of Gatheral et al. [2018] and the realized volatility estimators of Fukasawa [2021].

10.4 Multivariate extension

For a multivariate rough stochastic volatility model with assets $S^{(k)}$, $k = 1, \ldots, d$, and cross-variance swaps paying $\sum_i \Delta X_i^{(k)} \Delta X_i^{(\ell)}$, the discrete monitoring correction involves Malliavin derivatives $D_u V_s^{(k)}$ and their correlation structure. Under a factor model where all volatilities are driven by a common fractional kernel (as in multi-asset rough Bergomi), we conjecture that the cross-variance swap correction also scales as $O(n^{-(H+1/2)})$, with the coefficient depending on the cross-leverage structure. The multivariate case is particularly relevant for dispersion trading and correlation swap pricing.

10.5 Non-power-law kernels

Theorem 8 assumes $K(t) = t^{H-1/2}/\Gamma(H+\frac{1}{2})$. For general completely monotone kernels $K(t) = \int_0^\infty e^{-\lambda t}\,\mu(\mathrm{d}\lambda)$ (as in the Volterra Heston models of Abi Jaber et al. 2021), the key quantity is the behavior of $\int_0^\Delta K(\tau)\,\mathrm{d}\tau$ as $\Delta \to 0$. If $K(t) \sim c\,t^{\alpha-1}$ as $t \to 0$ for some $\alpha \in (0,1)$, then $\int_0^\Delta K \sim c\,\Delta^\alpha/\alpha$ and the discrete monitoring correction scales as $O(n^{-\alpha})$. The power-law case corresponds to $\alpha = H + \frac{1}{2}$. More generally, kernels with logarithmic corrections ($K(t) \sim t^{\alpha-1}\log(1/t)$) would produce $O(n^{-\alpha}\log n)$ corrections—an interesting direction that we do not pursue here.

11 Conclusion

We have shown that the discrete monitoring correction for variance swaps under Volterra stochastic volatility models with Hurst parameter $H \in (0, \frac{1}{2})$ scales as $O(n^{-(H+1/2)})$ when $\rho \neq 0$, a rate strictly slower than the classical $O(n^{-1})$. The mechanism is a leverage effect amplified by the rough kernel: the Malliavin derivative of the variance process inherits the singularity $K(s-u) \sim (s-u)^{H-1/2}$, producing a cross-term that dominates the standard convexity correction.

The result is universal within the Volterra SV class: the anomalous scaling rate depends only on the kernel exponent H and the leverage ρ, while the coefficient depends on the model through the single quantity $\bar{\varphi} = T^{-1}\int_0^T \mathbb{E}[\sigma_V(V_t)\sqrt{V_t}]\,\mathrm{d}t$. The rate is sharp (Corollary 10), the remainder is $o(n^{-1})$ uniformly with no phase transition (Proposition 12), and the boundary behavior connects naturally to the jump-model literature (Proposition 13).

Several extensions are discussed in Section 10. Among the most promising: (i) the rough Bergomi case (Conjecture 15), where the Gaussian Volterra structure requires a different remainder analysis; (ii) corridor variance swaps, whose monitoring-frequency dependence may yield a model-free estimator of the Hurst parameter H; (iii) the multivariate case, relevant to dispersion trading and correlation swap pricing; (iv) non-power-law kernels, where the short-time singularity of K determines the anomalous exponent.

From a methodological perspective, our work demonstrates that the Malliavin duality approach of Alòs and Fukasawa [2021] extends naturally to the Volterra setting. The key structural insight is that the anomalous scaling rate is determined entirely by the kernel singularity: any stochastic volatility model with $D_u V_s \sim (s-u)^{H-1/2}$ as $s \to u$ will produce the same $O(n^{-(H+1/2)})$ rate, regardless of the specific parametric form. This universality strengthens the practical relevance of the result, since model risk in the *rate* is limited to uncertainty in H, not in the full model specification.

A Malliavin derivative equation for Volterra SDE

Under Assumption (A2), the Malliavin derivative of V_t with respect to W_u satisfies the linear Volterra equation (cf. Alòs et al. 2007, Nualart 2006)

$$D_u V_t = K(t-u)\,\sigma_V(V_u) + \int_u^t K(t-s)\,b'(V_s)\,D_u V_s\,\mathrm{d}s + \int_u^t K(t-s)\,\sigma_V'(V_s)\,D_u V_s\,\mathrm{d}W_s. \quad (38)$$

For rough Heston ($b(v) = \theta - \kappa v$, $\sigma_V(v) = \nu\sqrt{v}$), this becomes

$$D_u V_t = K(t-u)\,\nu\sqrt{V_u} - \kappa \int_u^t K(t-s)\, D_u V_s \,\mathrm{d}s + \frac{\nu}{2}\int_u^t K(t-s)\,\frac{D_u V_s}{\sqrt{V_s}}\,\mathrm{d}W_s. \tag{39}$$

B Bound on the drift correction

We prove (16). Substituting the leading-order iterate $D_u V_s^{(0)} = K(s-u)\,\sigma_V(V_u)$ into the drift integral of (38):

$$\begin{aligned}\Big|\int_u^t K(t-s)\, b'(V_s)\, D_u V_s^{(0)}\,\mathrm{d}s\Big| &\le L_b\,\sigma_V(V_u)\int_u^t K(t-s)\,K(s-u)\,\mathrm{d}s \\ &= \frac{L_b\,\sigma_V(V_u)\,(t-u)^{2H}}{\Gamma(2H+1)},\end{aligned} \tag{40}$$

where the last equality uses the Beta-function convolution identity (6) and $|b'(v)| \le L_b$ (Assumption (A3)).

For rough Heston, $L_b = \kappa$ and $\sigma_V(V_u) = \nu\sqrt{V_u}$, giving $|R(u,s)| \le \kappa\nu\sqrt{V_u}\,(s-u)^{2H}/\Gamma(2H+1)$.

The relative magnitude of the correction is

$$\frac{|R(u,s)|}{K(s-u)\,\sigma_V(V_u)} \le \frac{L_b\,\Gamma(H+\frac{1}{2})}{\Gamma(2H+1)}\,(s-u)^{H+1/2},$$

which is bounded by $L_b\,\Gamma(H+\frac{1}{2})/\Gamma(2H+1)\cdot\Delta^{H+1/2}$ for $s \in [u, u+\Delta]$ and vanishes as $n \to \infty$.

C The stochastic correction vanishes in expectation

We show that the stochastic integral in (38) does not contribute to $\mathbb{E}[I_i\, M_i^W]$ at any order.

The stochastic correction to $D_u V_s$ is

$$S(u,s) = \int_u^s K(s-r)\,\sigma_V'(V_r)\, D_u V_r \,\mathrm{d}W_r.$$

This contributes to $\mathbb{E}[I_i M_i^W]$ through

$$\mathbb{E}\int_{t_i}^{t_i+\Delta}\Big(\int_u^{t_i+\Delta} S(u,s)\,\mathrm{d}s\Big)\sqrt{V_u}\,\mathrm{d}u.$$

For fixed u, the inner integral $\int_u^{t_i+\Delta} S(u,s)\,\mathrm{d}s$ involves a stochastic integral over $[u,s]$ with $s > u$. By Fubini's theorem for stochastic integrals, we may write this as an Itô integral over $[u, t_i+\Delta]$ with an adapted integrand.

Taking the conditional expectation given $\mathcal{F}_u$: since V_u is $\mathcal{F}_u$-measurable and the stochastic integral from u onward has zero conditional expectation (it is an Itô integral of an adapted, square-integrable process starting at time u), the tower property gives

$$\mathbb{E}\Big[\sqrt{V_u}\int_u^{t_i+\Delta} S(u,s)\,\mathrm{d}s\Big] = \mathbb{E}\Big[\sqrt{V_u}\,\mathbb{E}\Big[\int_u^{t_i+\Delta} S(u,s)\,\mathrm{d}s\,\Big|\,\mathcal{F}_u\Big]\Big] = 0.$$

This holds for each u, so the total contribution vanishes.

Remark 13. *This argument is valid for* any *Volterra SV model satisfying our standing assumptions—it requires only the martingale property of the Itô integral and the tower property of conditional expectation. No model-specific structure (affine, Gaussian, etc.) is used.*

D Propagation of the remainder to $\delta(n)$

We prove that $\sum_i |E_i| = O(T\, n^{-(2H+1)})$, where E_i is the contribution to $\mathbb{E}[I_i M_i^W]$ from the Malliavin derivative remainder $R(u,s)$.

From the bound (16) and the structure of the Clark–Ocone formula:

$$\begin{aligned}
|E_i| &\le \mathbb{E}\int_{t_i}^{t_i+\Delta}\Big(\int_u^{t_i+\Delta} |R(u,s)|\,\mathrm{d}s\Big)\sqrt{V_u}\,\mathrm{d}u \\
&\le \frac{L_b}{\Gamma(2H+1)}\mathbb{E}\int_{t_i}^{t_i+\Delta}\sigma_V(V_u)\sqrt{V_u}\,\frac{(t_i+\Delta-u)^{2H+1}}{2H+1}\,\mathrm{d}u \\
&\le \frac{L_b\,\bar\varphi_{\max}\,\Delta^{2H+2}}{(2H+1)(2H+2)\,\Gamma(2H+1)}.
\end{aligned}$$

Summing over n intervals:

$$\sum_{i=0}^{n-1}|E_i| \le \frac{n\,L_b\,\bar\varphi_{\max}\,\Delta^{2H+2}}{(2H+1)(2H+2)\,\Gamma(2H+1)} = \frac{L_b\,\bar\varphi_{\max}\,T^{2H+2}}{(2H+1)(2H+2)\,\Gamma(2H+1)\,n^{2H+1}}.$$

Dividing by T, the contribution to $\delta(n)$ is $O(T^{2H+1}\,n^{-(2H+1)})$. Since $2H+1>1$ for all $H>0$, this is always $o(n^{-1})$.

Remark 14 (Comparison with the loose bound)**.** *The loose bound replaces $\Delta^{2H+2} = \Delta^{3H+3/2}\cdot\Delta^{1/2-H}$ and drops $\Delta^{1/2-H}$, yielding $O(n^{-(3H+1/2)})$. Since $3H+\frac{1}{2} < 2H+1$ (equivalently $H<\frac{1}{2}$), the loose bound is strictly weaker. The discrepancy is most pronounced for small H: at $H=0.05$, the loose bound gives $O(n^{-0.65})$ while the tight bound gives $O(n^{-1.1})$—a difference of nearly half a unit in the exponent.*

E Malliavin differentiability

E.1 Rough Heston: localization argument

Assumption (A2) ($I_i \in \mathbb{D}^{1,2}$) is verified via a localization argument for the rough Heston dynamics.

The square-root coefficient $\nu\sqrt{V}$ fails to be globally Lipschitz, but is locally Lipschitz on $\{V>\epsilon\}$. Following Alòs et al. [2007], one introduces stopped processes $V_t^{(\epsilon)} = V_{t\wedge\tau_\epsilon}$ where $\tau_\epsilon = \inf\{t : V_t \le \epsilon\}$. Each $V_t^{(\epsilon)} \in \mathbb{D}^{1,2}$ by standard Malliavin theory for Volterra SDEs with Lipschitz coefficients [Nualart, 2006]. As $\epsilon\downarrow 0$, $V^{(\epsilon)} \to V$ in L^2 and $\tau_\epsilon \to \infty$ a.s. (since $V_t>0$ a.s. under the Feller-type condition for rough Heston), yielding $I_i \in \mathbb{D}^{1,2}$.

E.2 General Volterra SDE: sufficient conditions

For general Volterra SDE (5) with σ_V locally Lipschitz on $(0, \infty)$:

(i) If σ_V is globally Lipschitz (e.g., $\sigma_V(v) = \nu v^\gamma$ with $\gamma = 1$), then $V_t \in \mathbb{D}^{1,2}$ directly by Nualart [2006, Theorem 2.2.1] adapted to Volterra SDEs.

(ii) For $\sigma_V(v) = \nu v^\gamma$ with $\gamma \in [\frac{1}{2}, 1)$, the localization argument extends: on $\{V > \epsilon\}$, σ_V is Lipschitz with constant $\nu\gamma\epsilon^{\gamma-1}$, and $\tau_\epsilon \to \infty$ under appropriate moment conditions.

(iii) For $\gamma < \frac{1}{2}$ or σ_V with zeros, the localization argument requires additional care; we do not pursue this here.

E.3 Lifted Markovian route

An alternative approach works within the N-factor lifted Markovian approximation of Abi Jaber and El Euch [2019]. The approximation $V^{(N)}$ is a finite-dimensional diffusion, and standard Malliavin theory applies: $I_i^{(N)} \in \mathbb{D}^{1,2}$ for each N. The expansion (24) holds for each N, with $\bar{\varphi}^{(N)} \to \bar{\varphi}$ as $N \to \infty$ by the weak convergence results of Richard et al. [2023]. The result then follows by a standard approximation argument.

F Proof of the Beta convolution identity

We prove the identity (6): $\int_0^t K(t-s)\, K(s)\, \mathrm{d}s = t^{2H}/\Gamma(2H+1)$.

Proof. Write $K(r) = r^{H-1/2}/\Gamma(H + \frac{1}{2})$. Then

$$\int_0^t K(t-s)\, K(s)\, \mathrm{d}s = \frac{1}{\Gamma(H + \frac{1}{2})^2} \int_0^t (t-s)^{H-1/2}\, s^{H-1/2}\, \mathrm{d}s.$$

Substituting $s = t\, u$ with $\mathrm{d}s = t\, \mathrm{d}u$:

$$= \frac{t^{2H}}{\Gamma(H + \frac{1}{2})^2} \int_0^1 (1-u)^{H-1/2}\, u^{H-1/2}\, \mathrm{d}u = \frac{t^{2H}\, B(H+\frac{1}{2},\, H+\frac{1}{2})}{\Gamma(H + \frac{1}{2})^2},$$

where $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$ is the Beta function. Therefore

$$\int_0^t K(t-s)\, K(s)\, \mathrm{d}s = \frac{t^{2H}\, \Gamma(H + \frac{1}{2})^2}{\Gamma(H + \frac{1}{2})^2\, \Gamma(2H+1)} = \frac{t^{2H}}{\Gamma(2H+1)}.$$

□

More generally, for $\alpha, \beta > 0$:

$$\int_0^t \frac{(t-s)^{\alpha-1}}{\Gamma(\alpha)}\, \frac{s^{\beta-1}}{\Gamma(\beta)}\, \mathrm{d}s = \frac{t^{\alpha+\beta-1}}{\Gamma(\alpha + \beta)}. \tag{41}$$

This identity is used repeatedly in the Picard iteration analysis (with $\alpha = H + \frac{1}{2}$ and $\beta = H + \frac{1}{2}$, then $\beta = 2H + 1$, etc.).

G Convergence of the second Picard iterate

The second Picard iterate of the Malliavin derivative equation (38) produces a correction of the form

$$\begin{aligned} D_u V_t^{(2)} &= \int_u^t K(t-s)\, b'(V_s) \Big[-\frac{L_b\, \sigma_V(V_u)\, (s-u)^{2H}}{\Gamma(2H+1)} \Big] \mathrm{d}s \\ &= \frac{L_b^2\, \sigma_V(V_u)}{\Gamma(2H+1)} \int_u^t K(t-s)\, (s-u)^{2H}\, \mathrm{d}s \\ &= \frac{L_b^2\, \sigma_V(V_u)\, \Gamma(2H+1)\, (t-u)^{3H+1/2}}{\Gamma(2H+1)\, \Gamma(3H+\frac{3}{2})} = \frac{L_b^2\, \sigma_V(V_u)\, (t-u)^{3H+1/2}}{\Gamma(3H+\frac{3}{2})}. \end{aligned}$$

Here we used the convolution identity $\int_u^t K(t-s)(s-u)^{2H}\, \mathrm{d}s = \Gamma(2H+1)\, (t-u)^{3H+1/2} / (\Gamma(H+\frac{1}{2})\Gamma(2H+1) \cdot B(H+\frac{1}{2}, 2H+1))$. Simplifying via $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$:

$$\int_u^t K(t-s)(s-u)^{2H}\, \mathrm{d}s = \frac{\Gamma(2H+1)}{\Gamma(3H+\frac{3}{2})}\, (t-u)^{3H+1/2}.$$

The contribution to $\delta(n)$ is $O(\Delta^{3H+1/2+2}) = O(\Delta^{3H+5/2})$ per interval, summing to $O(n^{-(3H+3/2)})$. Since $3H + \frac{3}{2} > 2H + 1$ for all $H > 0$ (equivalently $H + \frac{1}{2} > 0$), the second iterate is always subdominant to the first.

The k-th iterate contributes $O(n^{-((k+1)H+k/2+1/2)})$, and the series converges geometrically with ratio $O(\Delta^{H+1/2}) \to 0$, confirming absolute convergence of the Picard expansion for all $H \in (0, \frac{1}{2})$.

Remark 15 (Convergence rate of the Picard series)**.** *The exponent of the k-th Picard iterate's contribution to $\delta(n)$ is $e_k = (k+1)H + k/2 + 1/2 = (k+1)(H+1/2) - 1/2 + 1/2 = (k+1)(H+1/2)$. Wait—let us recount. The k-th iterate of $D_u V_s$ has magnitude $O((s-u)^{(k+1)(H+1/2)-1})$ (from iterated convolution of K with itself). After inner integration (adding $+1$) and outer integration (adding $+1$), the contribution to $\mathbb{E}[I_i M_i^W]$ is $O(\Delta^{(k+1)(H+1/2)+1})$. Summing over n intervals: $O(n^{-((k+1)(H+1/2))})$.*

Therefore the contributions decay geometrically:

k	*Exponent of n*	*For $H = 0.1$*
0 (leading)	$H + 1/2 = 0.6$	$n^{-0.6}$
1 (first correction)	$2H + 1 = 1.2$	$n^{-1.2}$
2 (second correction)	$3H + 3/2 = 1.8$	$n^{-1.8}$
3	$4H + 2 = 2.4$	$n^{-2.4}$

Each successive term contributes at a rate faster by a factor of $n^{-(H+1/2)}$, confirming that the Picard series converges rapidly for the purpose of asymptotic analysis. The first correction ($k = 1$, exponent $2H + 1$) is the only one that could potentially compete with the convexity term $O(n^{-1})$; but since $2H + 1 > 1$ for all $H > 0$, even this is subdominant.

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